

Table 2 Dependence of heat transfer coefficient on surface roughness

Author(s)	Dependence	Pressure	Heat flux (W/m ²)	Fluid(s)	Heater type	Surface material	Surface preparation	Roughness (μm)
Stephan [28]	$h \propto R_{p,\text{old}}^{0.133}$	$P_r=0.023$	5×10^4	R-11	horizontal cylinder, horizontal plate	copper	not specified	$0.15 < R_{p,\text{old}} < 7.9$
Danilova and Bel'skii [29]	$h \propto R_z^{0.2}$	$0.118 < P_r < 0.155$	10^4	R-12, R-114	horizontal cylinder	cooper, steel	sandpaper, sandblasted, turning, etc.	$0.3 < R_z < 58$
Nishikawa et al. [30,31]	$h \propto R_{p,\text{old}}^{0.2(1-P_r)}$	$0.08 < P_r < 0.9$	10^5	R-12, R-113, R-114	horizontal plate	copper	emery cloth	$0.022 < R_{p,\text{old}} < 4.31$
Fedders [44]	$h \propto R_{p,\text{old}}^{0.133}$	$0.012 < P_r < 0.089$	5×10^4	water	horizontal cylinder	stainless steel	sandpaper, sandblasted	$0.18 < R_{p,\text{old}} < 3.6$
	$h \propto R_{p,\text{old}}^{0.12}$	$P_r=0.012$	10^6					
	$h \propto R_{p,\text{old}}^{0.10}$	$P_r=0.089$						
Ribatski and Jabardo [45]	$h \propto R_a^{0.2}$	$0.008 < P_r < 0.260$	not specified	R-11, R-123, R-12, R-134a, R-22	horizontal cylinder	copper, brass, stainless steel	sandpaper, sandblasted	$0.02 < R_a < 3.3$
Current work	$h \propto R_a^{0.1}$	$P_r=0.0046$	5×10^4 to 3×10^5	water	horizontal plate	aluminum	sandpaper, EDM	$0.038 < R_a < 10.0$
	$h \propto R_a^{0.2}$	$P_r=0.064$	2×10^4 to 8×10^4	FC-77				$0.027 < R_a < 10.0$